

ECE 210/211: Exam 1

Wednesday, February 19, 2025
7.15 - 8.30 p.m.

Name: (in BLOCK CAPITALS) _____

Solutions

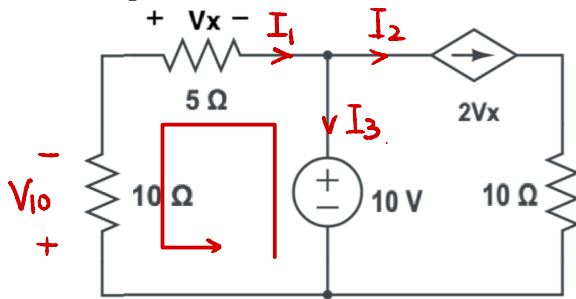
NetID: _____

Instructions

- Clearly PRINT your name in CAPITAL LETTERS.
- Clearly write your netID.
- This is a closed book and closed notes exam.
- Calculators and other electronic devices are not allowed.
- To get credit, you must SHOW ALL your work and provide explanations (used KCL, KVL, etc.).
- To get full credit, simplify your answers.
- All answers must INCLUDE UNITS whenever appropriate.
- Angles must be expressed in the range $(-\pi, \pi]$ rad.
- Write your final answers in the spaces provided or points may be deducted.
- The exam is printed double-sided.
- There is an empty page at the end of the exam in case you need additional space.

1. [28 points] The three parts of this problems are unrelated.

- (a) [10 pts] Consider the circuit below. Determine the power on the 10V voltage source using KCL and KVL.



Use SRS to mark current directions!

KCL @ top center node

$$I_1 = I_2 + I_3 = 2V_x + I_3$$

or use VDR

$$V_x = -10 \cdot \frac{5}{10+5} = -\frac{10}{3} \text{ V.}$$

↑
Vx and 10V source
has opposite polarity

KVL on left loop

$$10 + V_x + V_{10} = 0$$

$$10 + 5I_1 + 10I_1 = 0$$

$$\Rightarrow I_1 = -\frac{10}{15} = -\frac{2}{3} \text{ A}$$

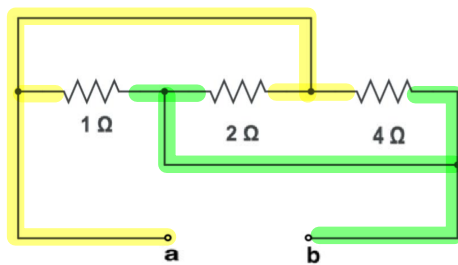
$$V_x = 5I_1 = -\frac{10}{3} \text{ V.}$$

$$I_3 = I_1 - 2V_x = -\frac{2}{3} - 2 \cdot \left(-\frac{10}{3}\right) = 6 \text{ A.}$$

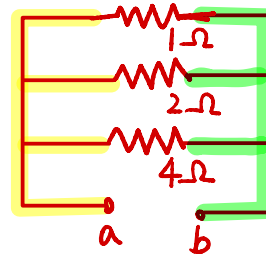
SRS! $P_{10V} = VI = 10 \cdot 6 = 60 \text{ W}$

$P_{10V} = \underline{60 \text{ W}}$

- (b) [6 pts] Determine R_{eq} between terminals a and b.



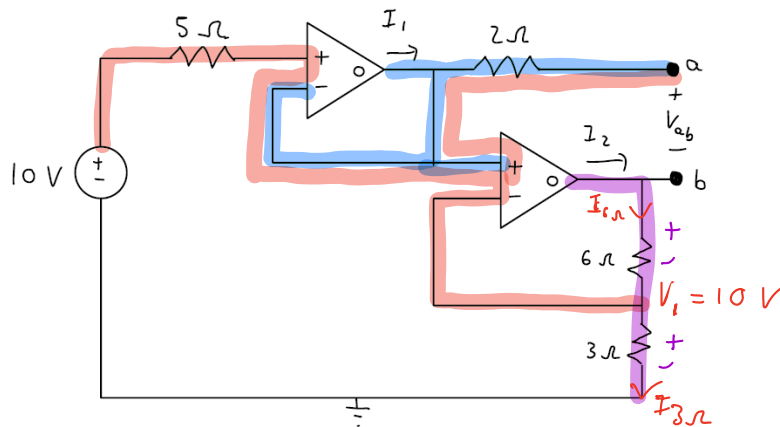
only two nodes



$$\frac{1}{R_{eq}} = \frac{1}{1} + \frac{1}{2} + \frac{1}{4} = \frac{7}{4}$$

$R_{eq} = \underline{\frac{4}{7} \Omega}$

- (c) [12 pts] Consider the circuit below, where the op-amps are operating in the linear regime under ideal op-amp conditions.



- i. [4 pts] Determine the current I_1 .

Following the blue path we see that $I_1 = 0$
 due to ideal op-amp $v_+ = v_- = 0$

$$I_1 = \underline{0 \text{ A}}$$

- ii. [8 pts] Determine the current I_2 and the voltage V_{ab} .

Following the red path we see that $v_i = 10 \text{ V} = v_a$
 due to ideal op-amp $v_+ = v_-$ & $i_+ = i_- = 0$.

$$\rightarrow I_{3\Omega} = \frac{v_i}{3} = \frac{10}{3} \text{ A} \rightarrow I_2 = I_{6\Omega} = I_{3\Omega} = \frac{10}{3} \text{ A}$$

$$V_b = V_{3\Omega} + V_{6\Omega} = I_3(3+6) = \frac{10}{3}(9) = 30 \text{ V}$$

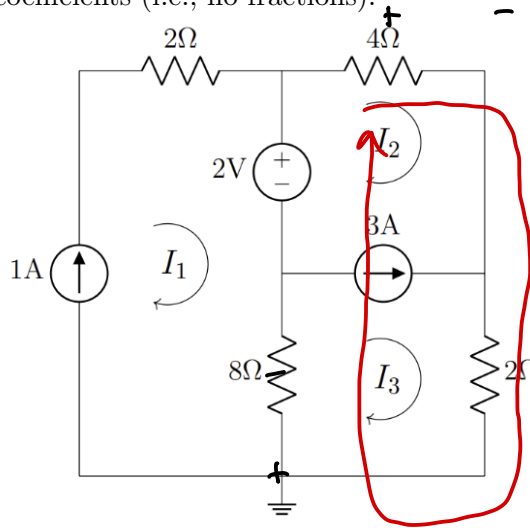
$$\rightarrow V_{ab} = V_a - V_b = 10 - 30 = -20 \text{ V}$$

$$I_2 = \underline{\frac{10}{3} \text{ A}}$$

$$V_{ab} = \underline{-20 \text{ V}}$$

2. [21 points] The two parts of this problem may be solved independently.

- (a) [9 pts] Use the loop-current method to obtain a set of three linearly independent equations in terms of the loop currents I_1 , I_2 , I_3 and no other variables, that can be used to determine the loop currents. Simplify your equations and write them with integer-valued coefficients (i.e., no fractions).



Loop I_1 :
 $I_1 = 1 \text{ A}$

Super loop:

$$2 = 4I_2 + 2I_3 + 8(I_3 - I_1)$$

$$2 = -8I_1 + 4I_2 + 10I_3$$

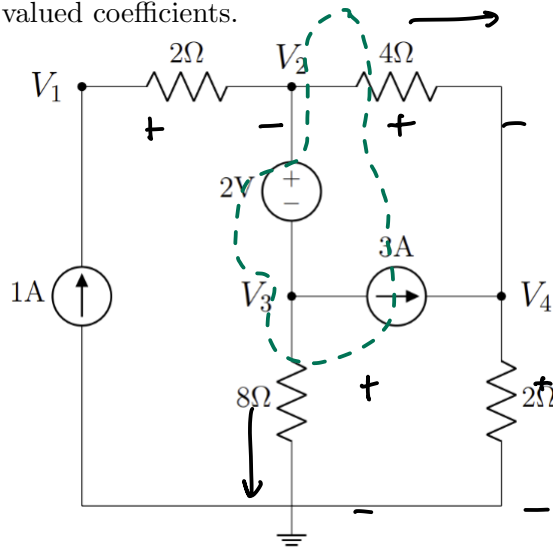
3A source
 $I_3 - I_2 = 3$

* Equations multiplied by
 -1 on both sides accepted.

$$\begin{array}{rcl} \underline{1} I_1 + \underline{0} I_2 + \underline{0} I_3 & = & \underline{1} \\ \underline{-8} I_1 + \underline{4} I_2 + \underline{10} I_3 & = & \underline{2} \\ \underline{0} I_1 + \underline{-1} I_2 + \underline{1} I_3 & = & \underline{3} \end{array}$$

3

- (b) [12 pts] Use the node-voltage method to obtain a set of four linearly independent equations in terms of the node voltages V_1, V_2, V_3, V_4 and no other variables, that can be used to determine the node voltages. Simplify your equations and write them with integer-valued coefficients.



$$\text{KCL @ } V_1$$

$$1 = \frac{V_1 - V_2}{2}$$

$$V_1 - V_2 = 2$$

$$\text{KCL @ Supernode}$$

$$\frac{V_1 - V_2}{2} = \frac{V_2 - V_4}{4} + \frac{V_3 - 0}{8} + 3$$

$$4V_1 - 4V_2 = 2V_2 - 2V_4 + V_3 + 24$$

$$4V_1 - 6V_2 - V_3 + 2V_4 = 24$$

$$\text{Supernode}$$

$$V_2 - V_3 = 2$$

$$\text{KCL @ } V_4$$

$$\frac{V_2 - V_4}{4} + 3 = \frac{V_4 - 0}{2}$$

$$V_2 - V_4 + 12 = 2V_4$$

$$V_2 - 3V_4 = -12$$

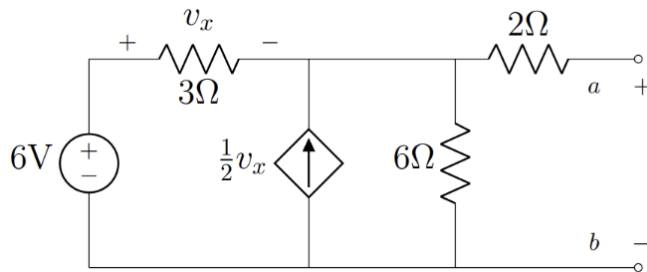
* Equations multiplied by -1 on both sides accepted.

* Another alternate equation obtained by KCL @ ground node.

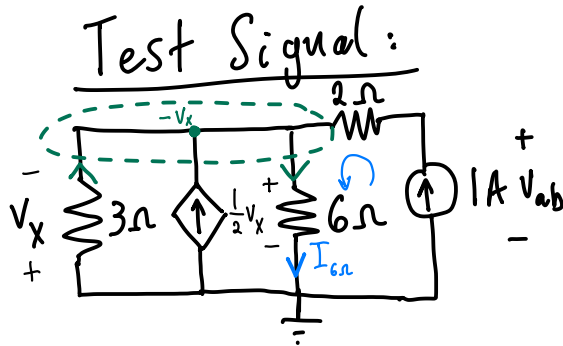
$$\begin{array}{rcl} \frac{1}{4} V_1 + \frac{-1}{-6} V_2 + \frac{0}{-1} V_3 + \frac{0}{2} V_4 & = & \frac{2}{24} \\ \frac{0}{0} V_1 + \frac{1}{1} V_2 + \frac{-1}{0} V_3 + \frac{0}{-3} V_4 & = & \frac{2}{-12} \end{array}$$

3. [25 points] The two parts of this problem may be solved independently.

(a) [13 pts] Consider the below circuit with open terminals as nodes a and b.



Determine the Thevenin resistance R_T as seen from nodes a and b.



KVL Loop:

$$V_{ab} = (2\Omega)(1A) - v_x$$

KCL:

$$\frac{v_x}{3} + \frac{1}{2}v_x + 1 = \frac{-v_x}{6}$$

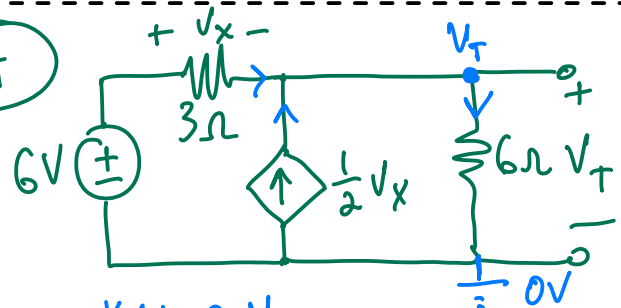
$$v_x = -1$$

* $V_{ab} = 2 + 1 = 3V$

$\rightarrow R_T = 3\Omega$

or find V_T and I_N , $R_T = \frac{V_T}{I_N}$:

V_T



KCL @ V_T

* $\frac{6 - V_T}{3} + \frac{1}{2}v_x = \frac{V_T}{6}$ * $V_x = 6 - V_T$

$$(6 - V_T)\left(\frac{5}{6}\right) = \frac{V_T}{6} \rightarrow \underline{V_T = 5V}$$

* $I_N = \frac{6 - v_x}{2}$

* KCL:

$$\frac{v_x}{3} + \frac{v_x}{2} = \frac{6 - v_x}{6} + \frac{6 - v_x}{2}$$

$\frac{3}{2}v_x = 4$

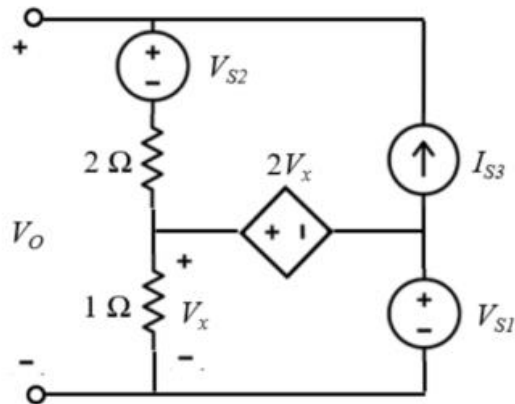
$v_x = \frac{8}{3}$

$\rightarrow \underline{I_N = \frac{5}{3}}$

$R_T = \underline{3\Omega}$

$$R_T = \frac{V_T}{I_N} = \frac{5}{\left(\frac{5}{3}\right)} = 3\Omega$$

(b) [12] Consider the resistive circuit below.



The voltage V_O can be expressed as:

$$V_O = K_1 V_{S1} + K_2 V_{S2} + K_3 I_{S3}$$

Use the superposition method to determine K_1 .

Solution:

Suppress (short) V_{S2} and (open) I_{S3} .

From KVL:

$$V_x = 2V_x + V_{S1} \text{ with } V_x = V_O$$

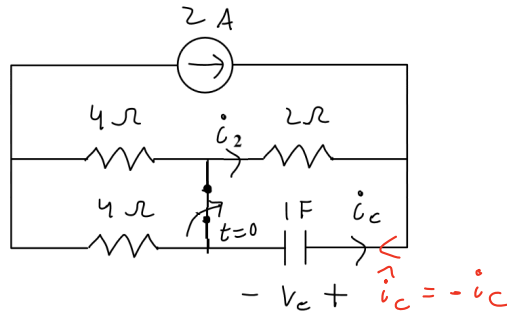
$$V_O = -V_{S1}$$

$$K_1 = -1$$

$$K_1 = \underline{\hspace{2cm}}$$

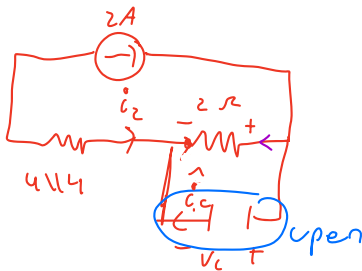
4. [26 points] The two parts of this problem may be solved independently.

- (a) [18 pts] Consider the circuit below, where the switch has been closed for a long time so the circuit is in steady state. The switch then opens at time $t = 0$ s.



Determine $v_c(t)$ and $i_2(t)$ for $t > 0$ s and sketch them for $t > -1$ s.

For $t < 0$, use circuit above in steady state:

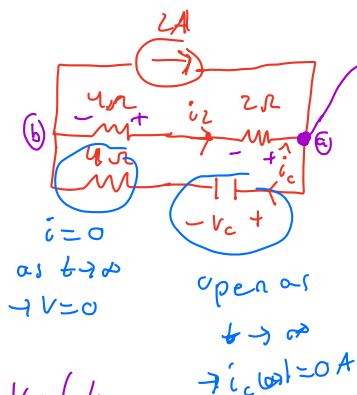


$$\text{KCL: } 2 + \hat{i}_2 = \hat{i}_c = 0 \rightarrow \hat{i}_2 = -2$$

$$i_2(0^-) = -2 \text{ A}$$

$$v_c(0^-) = v_{2\Omega} = -i_2(2) = 2(2) = 4 \text{ V} = v_c(0^-)$$

For $t > 0$



$$v_c(t) = k_1 + k_2 e^{-t/\tau}$$

$$\text{KCL: } 2 + \hat{i}_2 = \hat{i}_c \rightarrow \hat{i}_2 = \hat{i}_c - 2$$

$$v_c(\infty) = k_1 = v_{ab} = -i_2(4+2) = 2(6) = 12 \text{ V} = v_c(\infty)$$

$$v_c(0^+) = v_c(0^-) = k_1 + k_2 = 12 + k_2 \rightarrow k_2 = v_c(0^-) - 12 = 4 - 12 = -8 \text{ V}$$

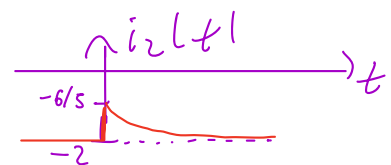
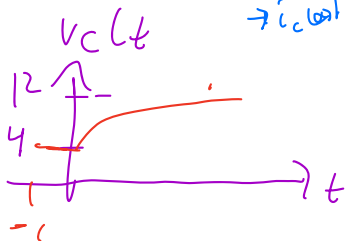
$$R_T = 4 + 4 + 2 = 10 \Omega = R_T$$

$$\tau = R_T C = (4 + 4 + 2) 1 = 10$$

$$\rightarrow v_c(t) = 12 - 8 e^{-t/10} \text{ V}$$

$$\rightarrow \hat{i}_c(t) = C \frac{dv}{dt} = (1) \left(\frac{8}{10} e^{-t/10} \right) = \frac{4}{5} e^{-t/10} \text{ A} = \hat{i}_c(t)$$

$$\rightarrow \hat{i}_2(t) = \hat{i}_c - 2 = \frac{4}{5} e^{-t/10} - 2 \text{ A}$$



$$v_c(t) = 12 - 8 e^{-t/10} \text{ V}$$

$$i_2(t) = \frac{4}{5} e^{-t/10} - 2 \text{ A}$$

remember to sketch

(b) [8 pts] Let $A = 1 - j\sqrt{3}$ and $B = \sqrt{2}e^{-j3\pi/4}$. Express A and $\frac{A}{B}$ in exponential form.

Solution:

$$A = 2e^{-j\frac{\pi}{3}}$$

$$\frac{A}{B} = \frac{2}{\sqrt{2}}e^{-j\frac{\pi}{3}+j\frac{3\pi}{4}} = \sqrt{2}e^{j\frac{5\pi}{12}}$$

$$A = \underline{\hspace{2cm}} \qquad \frac{A}{B} = \underline{\hspace{2cm}}$$